

Exp No: 1 (a)**Implementation of lexical analyzer using C programming****Aim:**

To implement a Lexical Analyzer using C programming that reads an input program and identifies different types of tokens such as keywords, identifiers, constants, operators, and special symbols.

Algorithm:

1. Start the program.
2. Read the input C program character by character.
3. Ignore spaces, tabs, and newlines.
4. If the character is an alphabet or `_`, collect characters until a non-alphanumeric character is found.
 - i. Check whether the collected word is a keyword.
 - ii. If not, identify it as an identifier.
5. If the character is a digit, collect all consecutive digits and identify them as a constant.
6. If the character is an operator such as `+`, `-`, `*`, `/`, `=`, `<`, `>`, identify it as an operator.
7. If the character is a special symbol such as `(`, `)`, `{`, `}`, `;`, `,`, identify it as a special symbol.
8. Continue until the end of the input.
9. Display all identified tokens.
10. Stop.

C Program:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
```

```
int isKeyword(char str[])
{
    char keywords[32][10] = {
        "auto", "break", "case", "char", "const", "continue",
        "default", "do", "double", "else", "enum", "extern",
        "float", "for", "goto", "if", "int", "long",
        "register", "return", "short", "signed", "sizeof", "static",
        "struct", "switch", "typedef", "union", "unsigned", "void",
        "volatile", "while"
    };

    int i;

    for (i = 0; i < 32; i++)
    {
        if (strcmp(str, keywords[i]) == 0)
            return 1;
    }

    return 0;
}
```

```

int main()
{
    char input[1000];
    int i = 0;

    printf("Enter the C program:\n");
    fgets(input, sizeof(input), stdin);

    printf("\nTokens:\n");

    while (input[i] != '\0')
    {
        /* Ignore spaces and newlines */
        if (isspace(input[i]))
        {
            i++;
        }

        /* Identifier or Keyword */
        else if (isalpha(input[i]) || input[i] == '_')
        {
            char word[50];
            int j = 0;

            while (isalnum(input[i]) || input[i] == '_')
            {
                word[j++] = input[i++];
            }

            word[j] = '\0';

            if (isKeyword(word))
                printf("%s -> Keyword\n", word);
            else
                printf("%s -> Identifier\n", word);
        }

        /* Constant */
        else if (isdigit(input[i]))
        {
            char number[50];
            int j = 0;

            while (isdigit(input[i]))
            {
                number[j++] = input[i++];
            }

            number[j] = '\0';

```

```

    printf("%s -> Constant\n", number);
}

/* Operators */
else if (input[i] == '+' || input[i] == '-' ||
        input[i] == '*' || input[i] == '/' ||
        input[i] == '=' || input[i] == '<' ||
        input[i] == '>')
{
    printf("%c -> Operator\n", input[i]);
    i++;
}

/* Special Symbols */
else if (input[i] == '(' || input[i] == ')' ||
        input[i] == '{' || input[i] == '}' ||
        input[i] == '[' || input[i] == ']' ||
        input[i] == ';' || input[i] == ',')
{
    printf("%c -> Special Symbol\n", input[i]);
    i++;
}

/* Unknown character */
else
{
    printf("%c -> Unknown Symbol\n", input[i]);
    i++;
}
}

return 0;
}

```

Execution Procedure

Step 1: Create the C File

Open **Notepad** and type/paste the above C program.

Save the file as:

lexer.c

While saving, select **Save as type** → **All Files**.

Make sure the file is saved as lexer.c and not lexer.c.txt.

Step 2: Open Command Prompt

Open Command Prompt and navigate to the folder where lexer.c is saved.

For example:

```
cd P:\pcd_lab\exp_1
```

Check whether the file is present:

```
ls
```

The file lexer.c should be displayed.

Step 3: Compile the C Program

Type:

```
gcc lexer.c -o lexer.exe
```

If there is no error, the executable file lexer.exe will be created.

Step 4: Run the Program

Type:

```
lexer.exe
```

The program will ask:

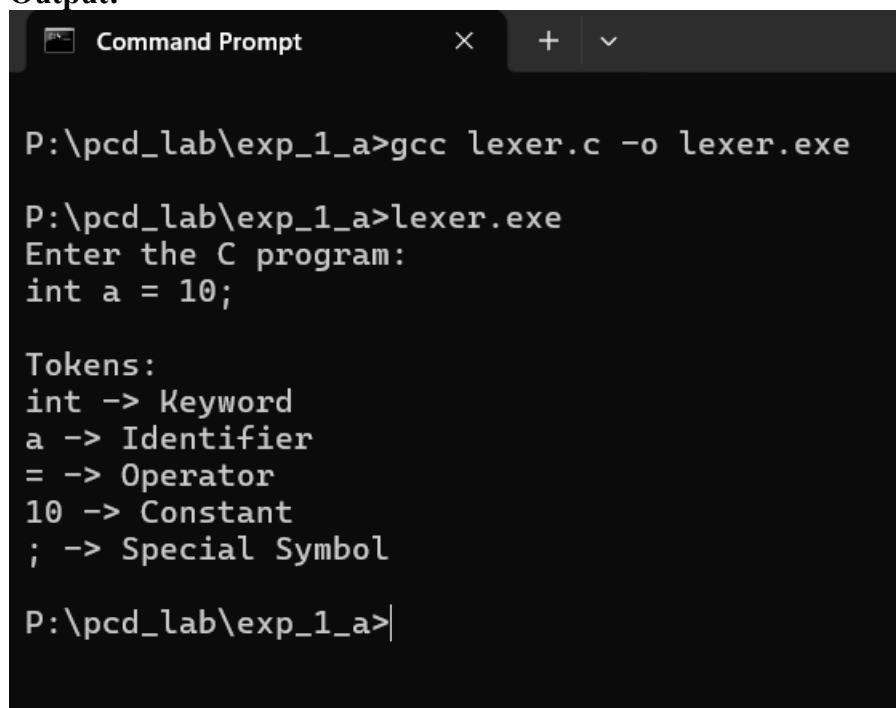
Enter the C program:

Enter the following input:

```
int a = 10;
```

Press **Enter**.

Output:



```
P:\pcd_lab\exp_1_a>gcc lexer.c -o lexer.exe

P:\pcd_lab\exp_1_a>lexer.exe
Enter the C program:
int a = 10;

Tokens:
int -> Keyword
a -> Identifier
= -> Operator
10 -> Constant
; -> Special Symbol

P:\pcd_lab\exp_1_a>
```

Result:

Thus, the Lexical Analyzer was successfully implemented in C, and the input program was successfully divided into different tokens such as **keywords, identifiers, constants, operators, and special symbols**.